

Covariance and Correlation Exercises

Mathematics for AI

Question 1. Covariance and correlation from observed data

Let X and Y be discrete random variables. After observing (X, Y) six times, we record the following outcomes:

$(1, 5), (4, 18), (3, 17), (8, 35), (1, 6), (3, 20).$

i	x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$
1	1	5			
2	4	18			
3	3	17			
4	8	35			
5	1	6			
6	3	20			
Total					

Set-up:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

(a)

Calculate $\text{Cov}(X, Y)$.

$$\begin{aligned} f_X(x) &= \sum_y f_{X,Y}(x, y) \\ \mathbb{E}[X] &= \sum_x x f_X(x) \\ &= \frac{1}{6} \sum_{i=1}^6 x_i \\ &= \frac{1}{6} (1 + 4 + 3 + 8 + 1 + 3) \\ &= \frac{20}{6} = \frac{10}{3} \end{aligned}$$

$$\begin{aligned}
f_Y(y) &= \sum_x f_{X,Y}(x, y) \\
\mathbb{E}[Y] &= \sum_y y f_Y(y) \\
&= \frac{1}{6} \sum_{i=1}^6 y_i \\
&= \frac{1}{6} (5 + 18 + 17 + 35 + 6 + 20) \\
&= \frac{101}{6}
\end{aligned}$$

i	x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$
1	1	5	$-\frac{7}{3}$	$-\frac{71}{6}$	$\frac{497}{18}$
2	4	18	$\frac{2}{3}$	$\frac{7}{6}$	$\frac{14}{18}$
3	3	17	$-\frac{1}{3}$	$\frac{1}{6}$	$-\frac{1}{18}$
4	8	35	$\frac{14}{3}$	$\frac{109}{6}$	$\frac{1526}{18}$
5	1	6	$-\frac{7}{3}$	$-\frac{65}{6}$	$\frac{455}{18}$
6	3	20	$-\frac{1}{3}$	$\frac{19}{6}$	$-\frac{19}{18}$
Total			0	0	$\frac{2472}{18}$

Now we need to find $\text{Cov}(X, Y)$. We know the following for any function of g of two random variables:

$$\mathbb{E}[g(X, Y)] = \sum_x \sum_y g(x, y) f_{X,Y}(x, y)$$

So then we can work out $\text{Cov}(X, Y)$, as follows:

$$\text{Cov}(X, Y) = \sum_x \sum_y (x - \mathbb{E}[X])(y - \mathbb{E}[Y]) f_{X,Y}(x, y)$$

Note $\sum_x \sum_y$ means for each of the possible x values consider every possible y value/ Our x values are $\{1, 3, 4, 8\}$ and our y values are $\{5, 6, 17, 18, 20, 35\}$, so the grid has $4 \times 6 = 24$ cells. But only 6 of these cells were outcomes we observed. The other 18 have $f_{X,Y}(x, y) = 0$, so their terms amount to 0 as you are multiplying by 0 so we can simplify to:

$$\begin{aligned}
\text{Cov}(X, Y) &= \sum_x \sum_y (x - \mathbb{E}[X])(y - \mathbb{E}[Y]) f_{X,Y}(x, y) \\
&= \frac{1}{6} \sum_{i=1}^6 (x_i - \mathbb{E}[X])(y_i - \mathbb{E}[Y]) \\
&= \frac{1}{6} \cdot \frac{2472}{18} \\
&= \frac{2472}{108} = \frac{206}{9}
\end{aligned}$$

(b)

Calculate $\rho(X, Y)$.

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[(X - \mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2 - 2X \mathbb{E}[X] + (\mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2] - 2\mathbb{E}[X] \mathbb{E}[X] + (\mathbb{E}[X])^2 \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2\end{aligned}$$

We have $\mathbb{E}[X]$ but we don't have $\mathbb{E}[X^2]$, so we need to work this out:

$$\begin{aligned}f_X(x) &= \sum_y f_{X,Y}(x, y) \\ \mathbb{E}[X^2] &= \sum_x x^2 f_X(x) \\ &= \frac{1}{6} \sum_{i=1}^6 (x_i)^2 \\ &= \frac{1}{6} (1 + 16 + 9 + 64 + 1 + 9) \\ &= \frac{100}{6} = \frac{50}{3}\end{aligned}$$

Now we can calculate $\text{Var}(X)$:

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\ &= \frac{50}{3} - \left(\frac{10}{3}\right)^2 \\ &= \frac{50}{3} - \frac{100}{9} \\ &= \frac{150 - 100}{9} = \frac{50}{9}\end{aligned}$$

So now we can calculate the σ_X :

$$\begin{aligned}\sigma_X &= \sqrt{\text{Var}(X)} \\ &= \sqrt{\frac{50}{9}}\end{aligned}$$

Similarly we need to calculate $E[Y^2]$:

$$\begin{aligned}
 f_Y(y) &= \sum_x f_{X,Y}(x, y) \\
 \mathbb{E}[Y^2] &= \sum_y y^2 f_Y(y) \\
 &= \frac{1}{6} \sum_{i=1}^6 (y_i)^2 \\
 &= \frac{1}{6} (25 + 324 + 289 + 1225 + 36 + 400) \\
 &= \frac{2299}{6}
 \end{aligned}$$

Now we can calculate $\text{Var}[Y]$:

$$\begin{aligned}
 \text{Var}(Y) &= \mathbb{E}[Y^2] - (\mathbb{E}[Y])^2 \\
 &= \frac{2299}{6} - \left(\frac{101}{6}\right)^2 \\
 &= \frac{2299}{6} - \frac{10201}{36} \\
 &= \frac{13794 - 10201}{36} = \frac{3593}{36}
 \end{aligned}$$

So now we can calculate the σ_Y :

$$\begin{aligned}
 \sigma_Y &= \sqrt{\text{Var}(Y)} \\
 &= \sqrt{\frac{3593}{36}}
 \end{aligned}$$

$$\begin{aligned}
 \rho(X, Y) &= \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \\
 &= \frac{\frac{206}{9}}{\sqrt{\frac{179650}{324}}} \\
 &= \frac{\frac{206}{9}}{5\sqrt{\frac{7186}{18}}} \approx 0.972
 \end{aligned}$$

Interpretation: Considering $\rho(X, Y)$ firstly our value is valid ($-1 \leq \rho \leq 1$). Now considering just the value, it is very close to 1 and then the sign is positive so this indicates that X and Y are very strongly linearly correlated. So a larger value of X would very likely be accompanied by a larger value of Y .

Interpretation:

The value of $\rho(X, Y) \approx 0.97$ is positive and very close to 1, which indicates a strong positive linear relationship between X and Y : larger values of X tend to occur alongside larger values of

Y . Note that $-1 \leq \rho \leq 1$ always holds, so this value is admissible, and unlike $\text{Cov}(X, Y) = \frac{206}{9}$, which carries the units of X and Y , the correlation is dimensionless and therefore comparable across datasets.